RECOMMENDATION SYSTEM - 5

* Agenda →

interactions matrix

Rating

User	Avengers	Titanic	Conjuring	Toy Story
Alice	5	2	?	4
Bob	4	?	5	2
Charlie	?	5	1	4

* Recap

→ code

Scratch

MF paper →

MF

IMDb

Netflix

1 lac users
17000 movies

0.1%

99.9%

missing

Sparse matrix

MF

y_i → latent factors

Action	Sci-fi

4k

d = 2, d = 10, d = 100

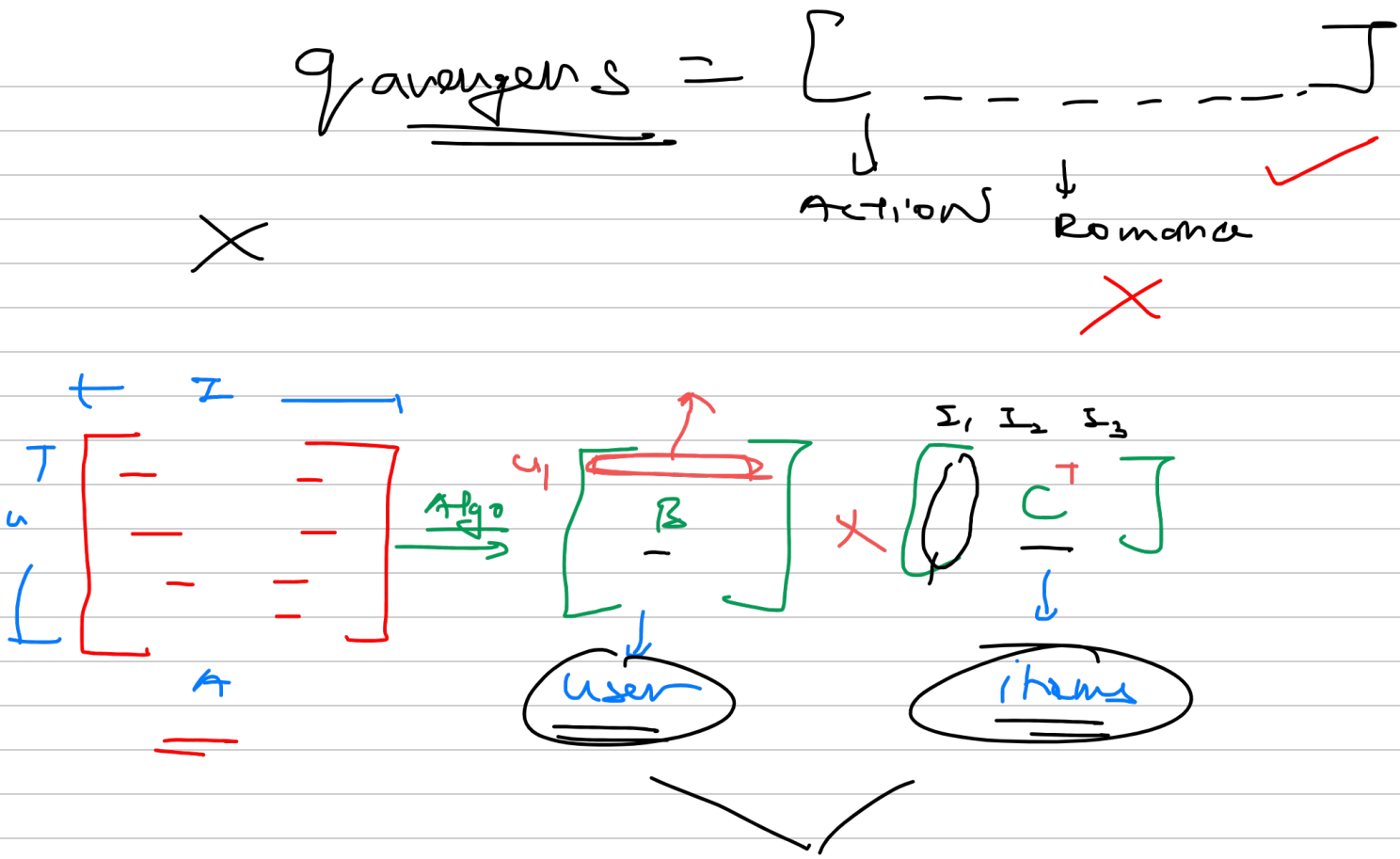
Train, Test ✓

Alice → (rating 17000 movies)

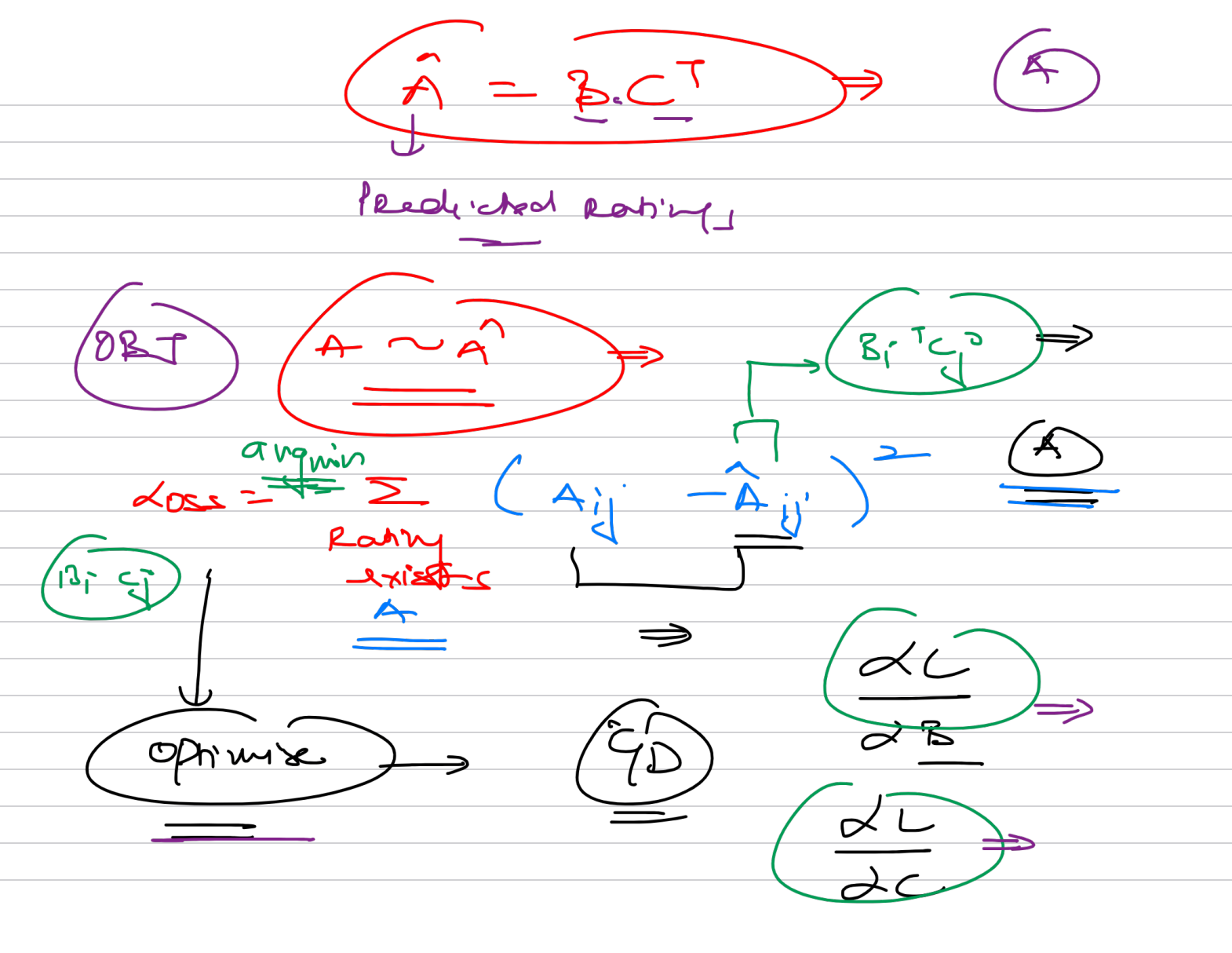
50 no.

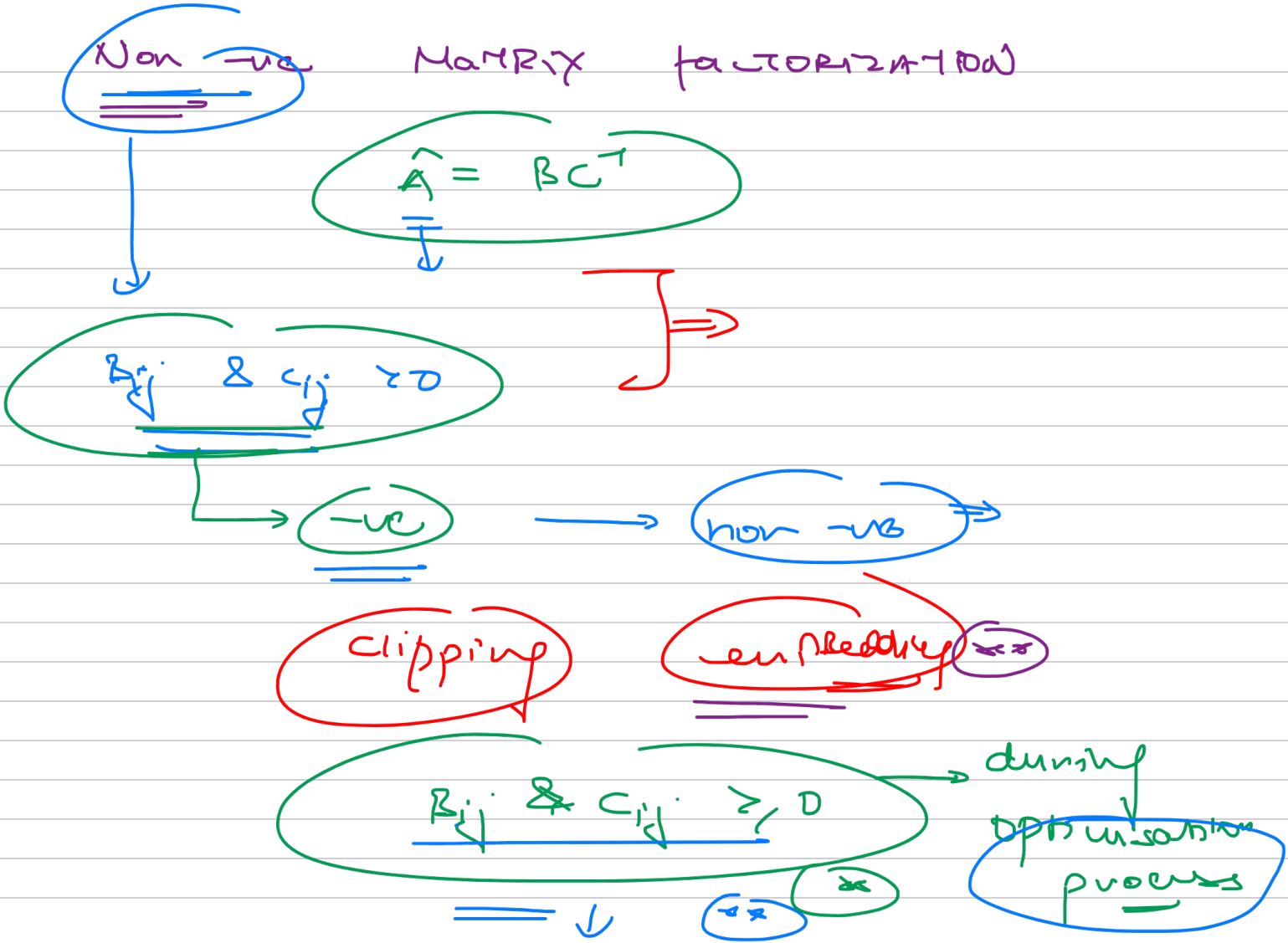
d = 50

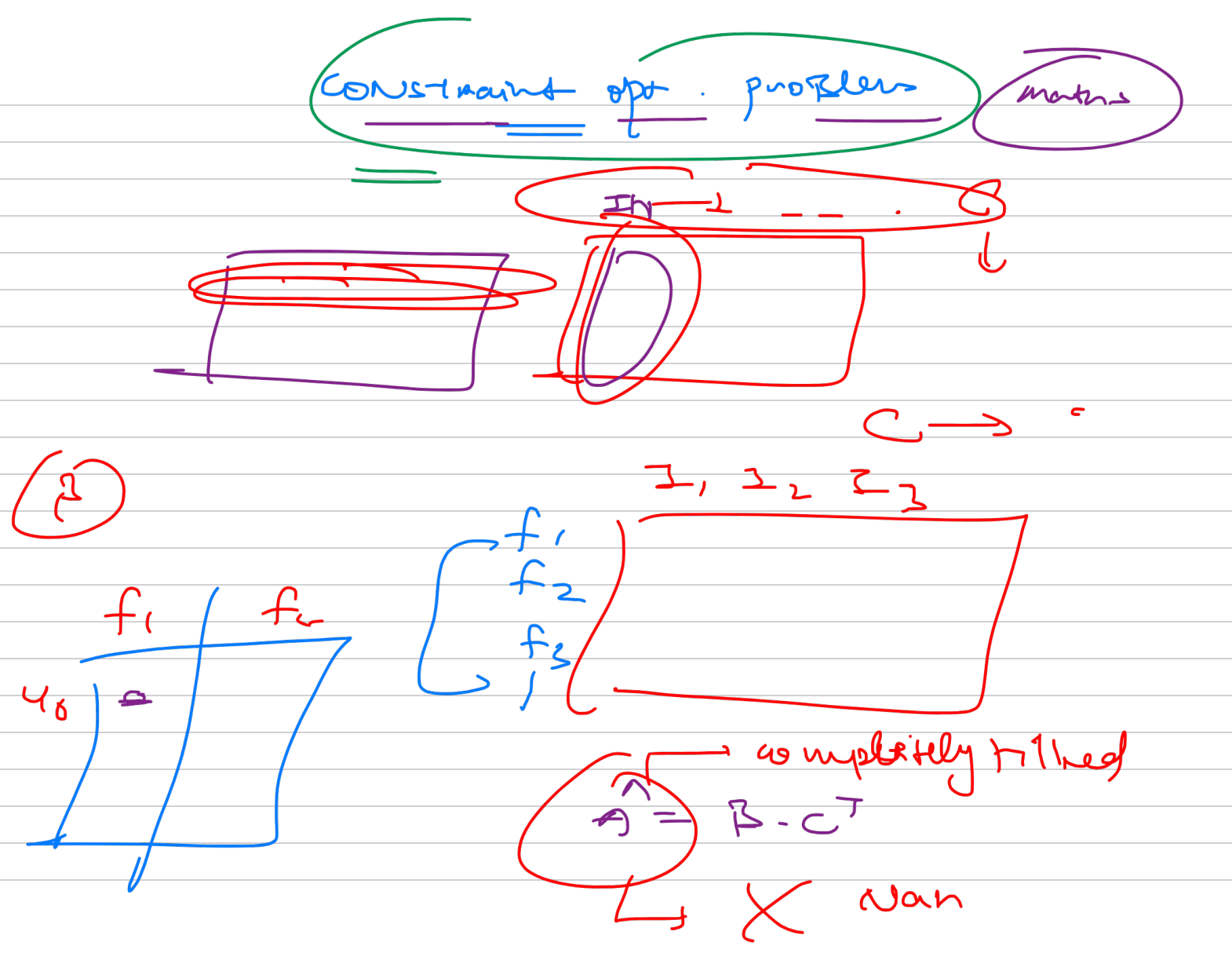
[latens]

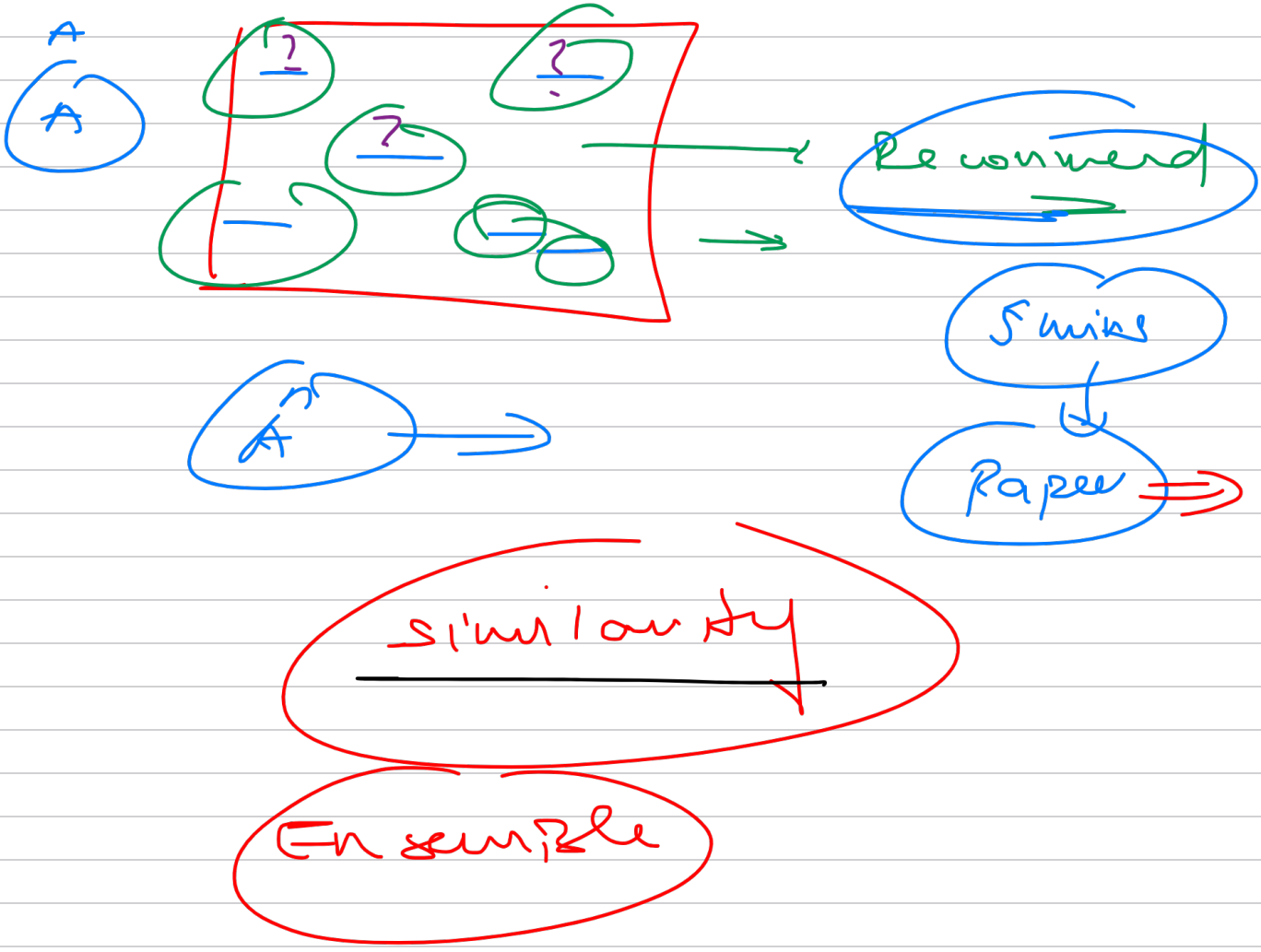


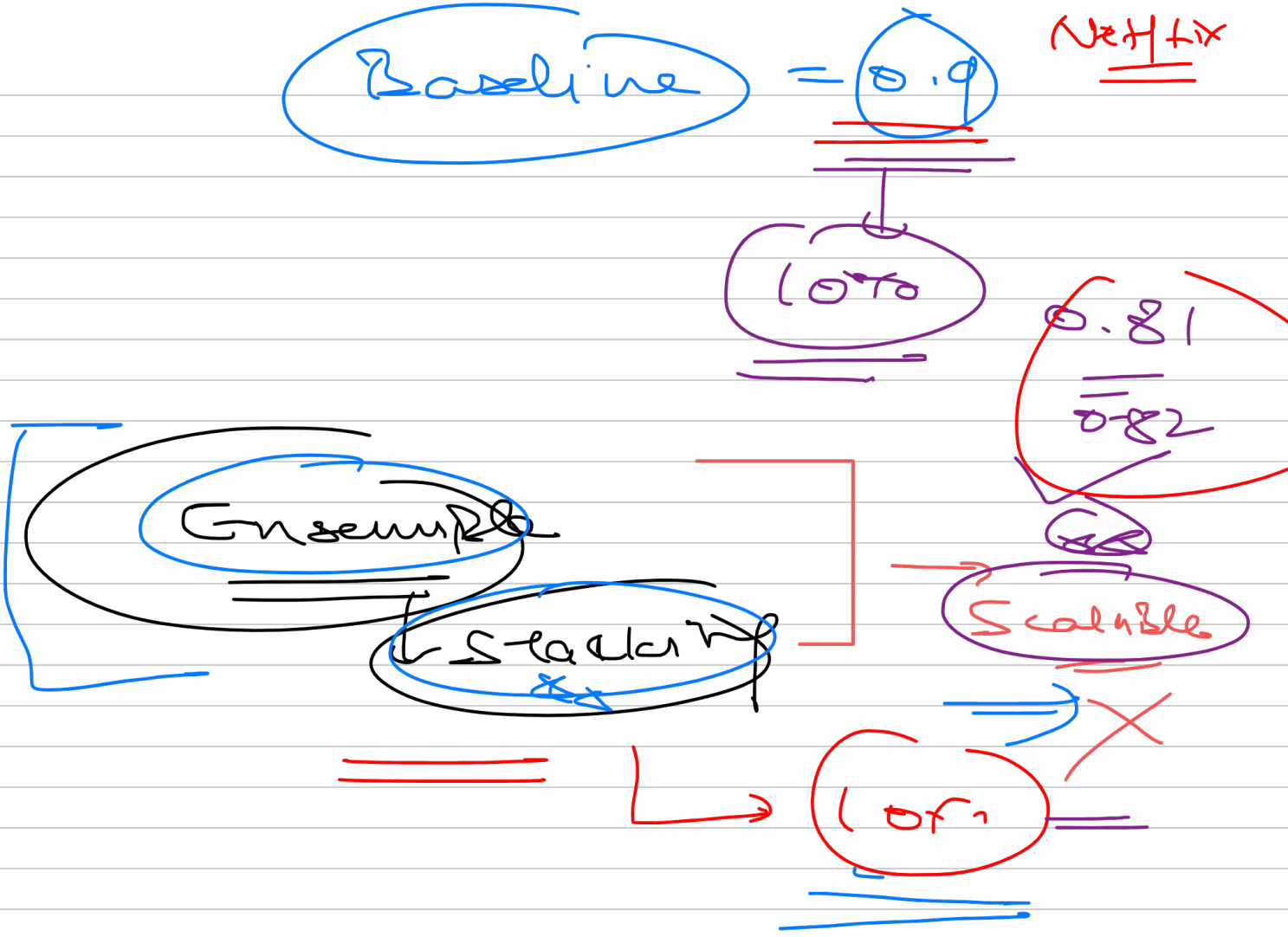
1) Randomly init A & C
↓ optimisation


$\hat{A} = B \cdot C^T$
A
Predicted Rating
OBJ
$A \sim \hat{A}$
$B_i^T C_j$
A
$\text{Loss} = \sum_{\text{argmin}}$
Rating exists
A
$(A_{ij} - \hat{A}_{ij})^2$
B_i C_j
Optimize
GD
$\frac{\partial L}{\partial B}$
$\frac{\partial L}{\partial C}$


Non -ve
MATRIX
FACTORIZATION
$\hat{A} = BC^T$
B_{ij} & $C_{ij} \geq 0$
-ve
non -ve
clipping
enabled
B_{ij} & $C_{ij} \geq 0$
during
optimisation
process


CONSTRAINT OPT. PROBLEMS
matrix
C →
f_1
f_2
f_3
f_1
f_2
y_0
completely filled
$A = B \cdot C^T$
X
Nan


A
A
?
?
?
Recommend
Swiss
Papee
A'
similarity
Ensemble


Baseline
= 0.9
Netflix
Lotto
0.81
0.82
Ensemble
Stacking
Scalable
Lotto

Yahoo Research → MATMIX

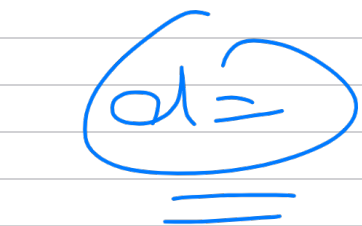
Item

100 Mn

User

GSM → Provisioning

SAM → Validation



$$\hat{z}_{ui} = q_i^T p_u$$

2 dim $\rightarrow$ f_1 = Action, f_2 = Romance

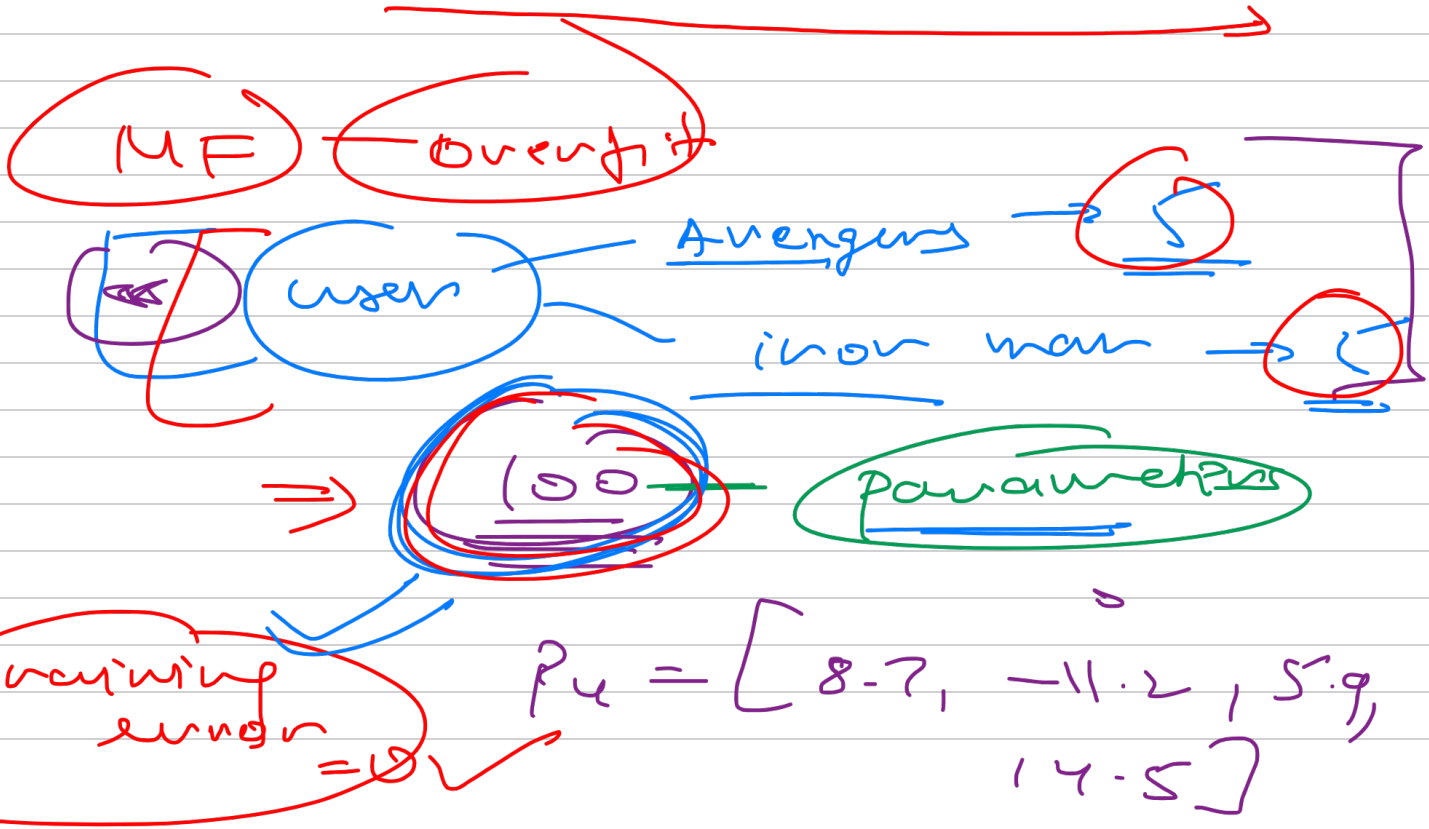
Alice =

$$p_A = \begin{bmatrix} 0.9 & 0.2 \end{bmatrix}$$

$$q_{Aug} = \begin{bmatrix} 0.95 & 0.1 \end{bmatrix}$$

$p_A \cdot q_{Aug}$ $\uparrow$

$d = 2, 10, 15, 20, 100, 1500$



Text → ↓

Regularization

2 models

Simpler ones

Acom Razor

Polynomial → 100 → ↑ → 95%

→ 2 → 95%

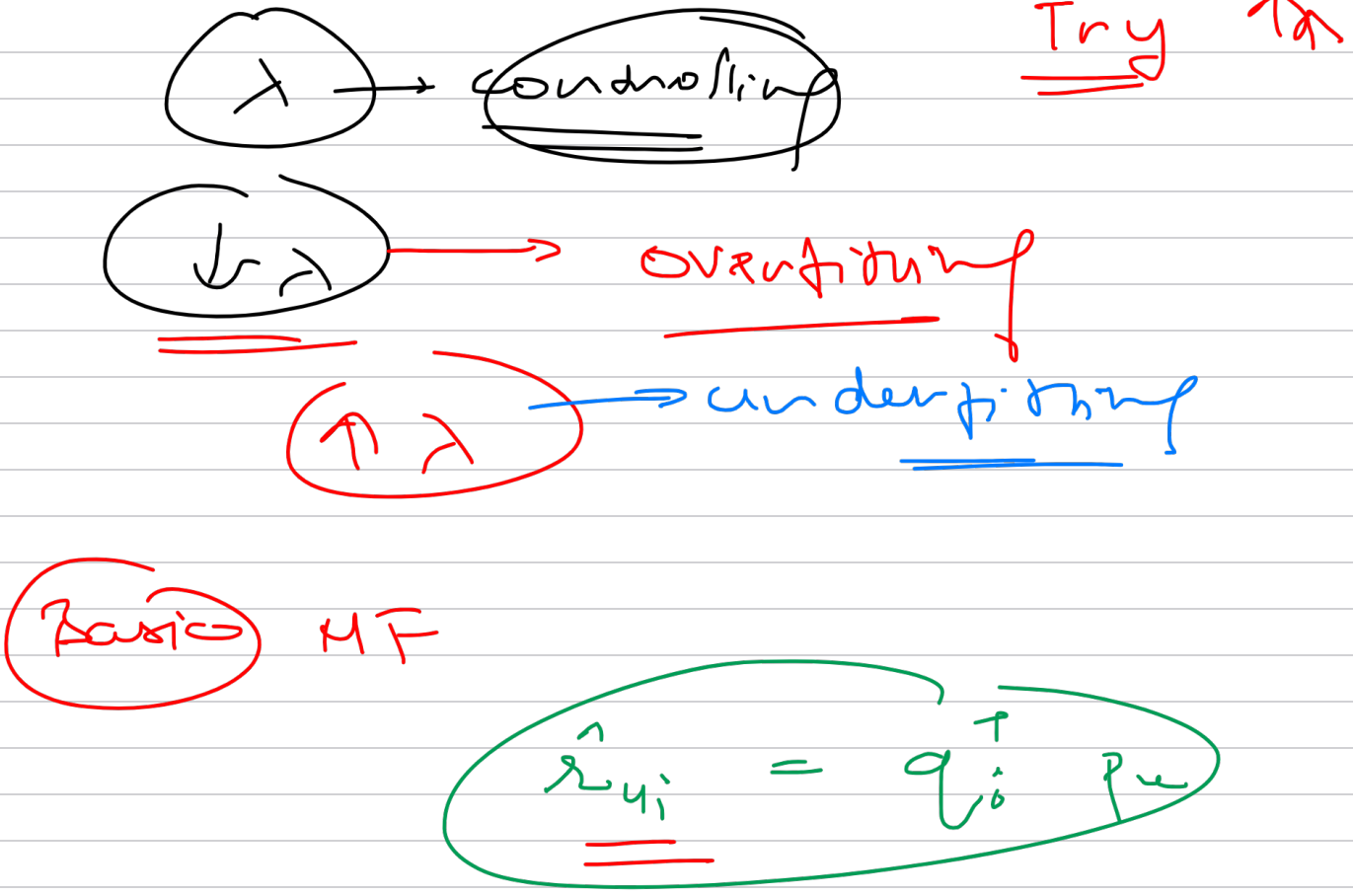
Model A → 0.01

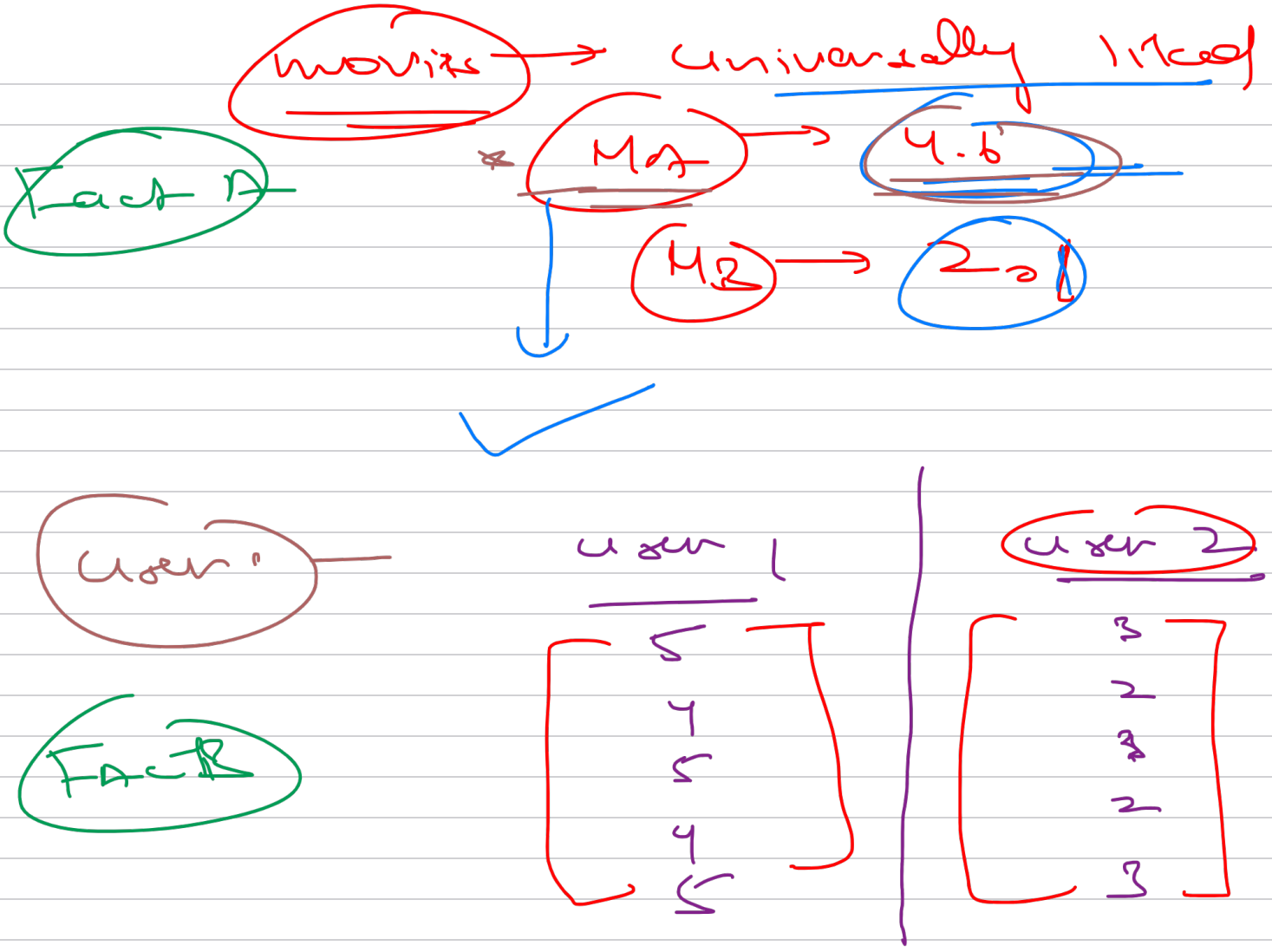
→ [15, −22, 11, 19, ...]

Model B → 0.02

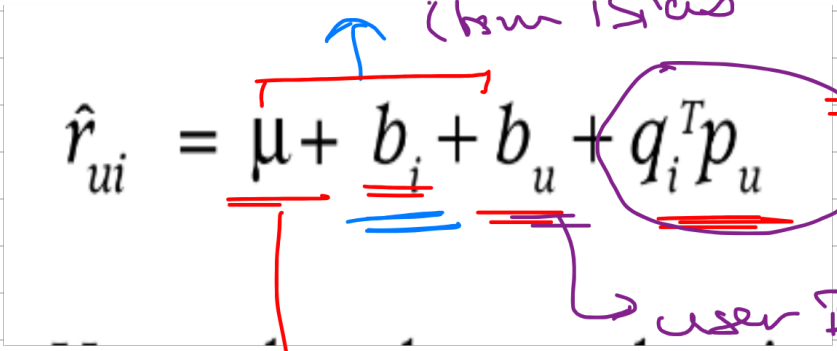
generalize better

[0.8, −0.5, 1.1, −0.3, ...]


λ
controlling
Try
↓ λ
overfitting
↑ λ
underfitting
Basic MF
$\hat{r}_{ui} = q_i^T p_u$


movies
universally liked
Fact A
MA
4.6
MB
2.1
User
Fact B
user 1
5
4
5
4
5
user 2
3
2
3
2
3

Bias model



$$\hat{r}_{ui} = \mu + b_i + b_u + q_i^T p_u$$

Titanic

movie

$\mu = 3.7$

$b_{Titanic} = +0.5$

3
2
1

For

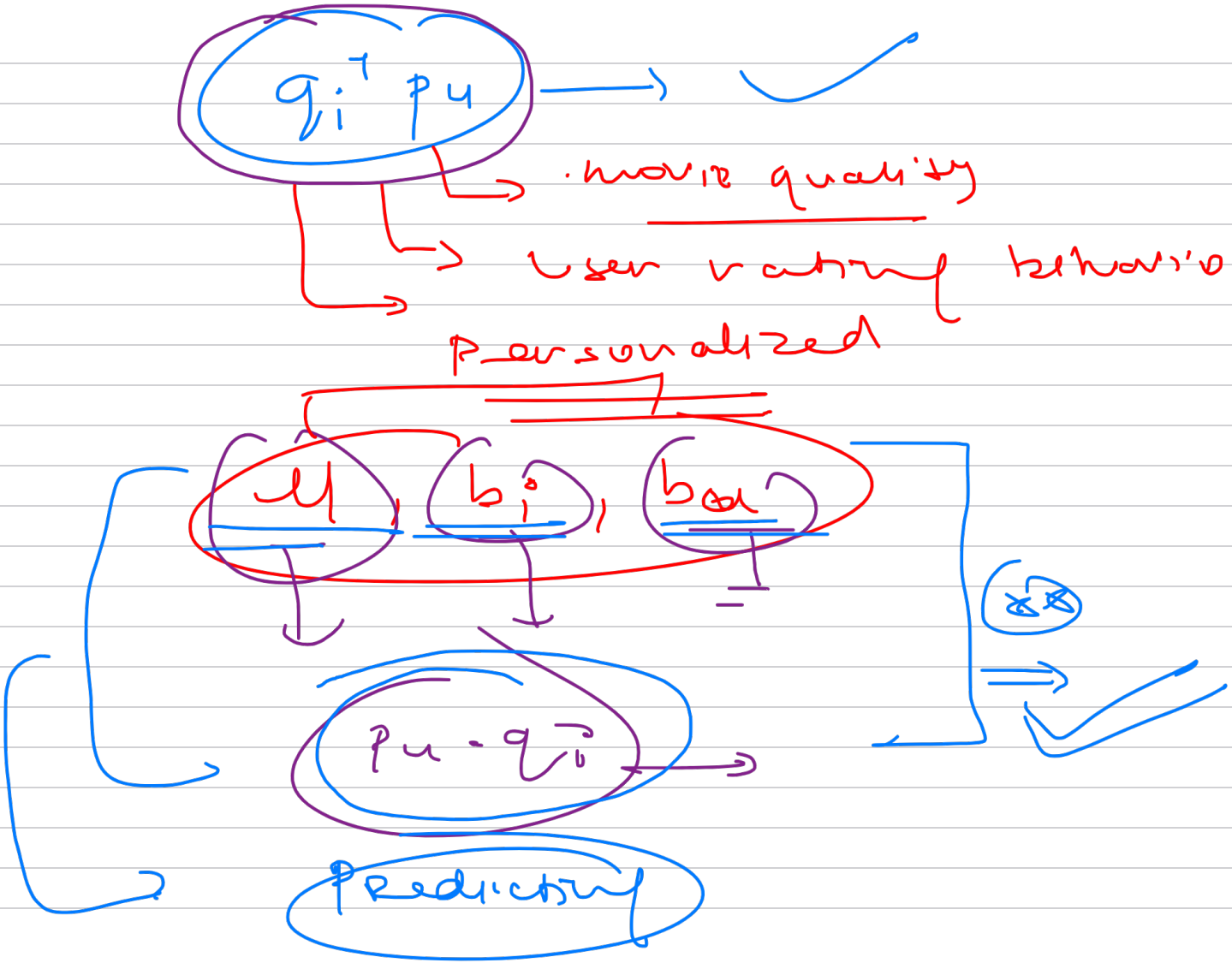
-0.3

-0.3

$3.7 + 0.5 - 0.3$

$= 3.9$

Titanic


$q_i^T p_u$
movie quality
user rating behavior
personalized
u
b_i
b_{ui}
$p_u \cdot q_i$
Predicting

global eq = 3.5

oppenheimer ⇒ + 0.5

gen eq = +0.3

4.5 ✓ = 4.3

✗ Alice ⇒ long

$P_A^{\uparrow} q_0 = -0.8$

4.3 − 0.8 = 3.5